Exercise 10.1.1 Show the following

(1)

$$[\Omega_1^{(1)} \otimes I^{(2)}, I^{(1)} \otimes \Lambda_2^{(2)}] = 0 \text{ for any } \Omega_1^{(1)} \text{ and } \Lambda_2^{(2)}$$

We determine what this commutator's action is on a test direct product of kets $|\psi\rangle = |a\rangle \otimes |b\rangle$. We evaluate to see that the action of this commutator on any vector in vector space $V_1 \otimes V_2$ is equal to zero, thus the commutator equals zero. We make use of (10.1.14) to resolve the direct product of two operators and their action on product kets.

$$\begin{aligned}
&\left[\Omega_1^{(1)} \otimes I^{(2)}, I^{(1)} \otimes \Lambda_2^{(2)}\right] |\psi\rangle = \\
&= \big((\Omega_1^{(1)} \otimes I^{(2)})(I^{(1)} \otimes \Lambda_2^{(2)}) - (I^{(1)} \otimes \Lambda_2^{(2)})(\Omega_1^{(1)} \otimes I^{(2)})\big)(|a\rangle \otimes |b\rangle) \\
&= (\Omega_1^{(1)} \otimes I^{(2)})(|I^{(1)}a\rangle \otimes |\Lambda_2^{(2)}b\rangle) - (I^{(1)} \otimes \Lambda_2^{(2)})(|\Omega_1^{(1)}a\rangle \otimes |I^{(2)}b\rangle) \\
&= |\Omega_1^{(1)}a\rangle \otimes |\Lambda_2^{(2)}b\rangle - |\Omega_1^{(1)}a\rangle \otimes |\Lambda_2^{(2)}b\rangle = 0
\end{aligned}$$

(operators of particle 1 commute with those of particle 2).

(2) Show that,

$$(\Omega_1^{(1)} \otimes \Gamma^{(2)})(\Theta^{(1)} \otimes \Lambda_2^{(2)}) = (\Omega\Theta)_1^{(1)} \otimes (\Gamma\Lambda)_2^{(2)}$$

We verify this identity by, again, acting on the test product ket $|\psi\rangle = |a\rangle \otimes |b\rangle$

$$\begin{aligned}
(\Omega_1^{(1)} \otimes \Gamma^{(2)})(\Theta^{(1)} \otimes \Lambda_2^{(2)}) |\psi\rangle &= (\Omega_1^{(1)} \otimes \Gamma^{(2)})(|\Theta^{(1)}a\rangle \otimes |\Lambda_2^{(2)}b\rangle) \\
&= |\Omega_1^{(1)}\Theta^{(1)}a\rangle \otimes |\Gamma^{(2)}\Lambda_2^{(2)}b\rangle \\
&= ((\Omega\Theta)_1^{(1)} \otimes (\Gamma\Lambda)_2^{(2)}) |\psi\rangle
\end{aligned}$$

(3) Show that if $[\Omega_1^{(1)}, \Lambda_1^{(1)}] = \Gamma_1^{(1)}$ then $[\Omega_1^{(1)\otimes(2)}, \Gamma_1^{(1)\otimes(2)}] = \Gamma_1^{(1)} \otimes I^{(2)}$

We start with the right hand side and expand the commutator.

$$\begin{aligned}
\Omega_1^{(1)\otimes(2)}\Gamma_1^{(1)\otimes(2)} - \Gamma_1^{(1)\otimes(2)}\Omega_1^{(1)\otimes(2)} &= \Omega_1^{(1)}\Gamma_1^{(1)} \otimes \mathbb{1}^{(2)}\mathbb{1}^{(2)} - \Gamma_1^{(1)}\Omega_1^{(1)} \otimes \mathbb{1}^{(2)}\mathbb{1}^{(2)} \\
&= \Omega_1^{(1)}, \Gamma_1^{(1)}] \otimes \mathbb{1}^{(2)} = \Gamma_1^{(1)} \otimes \mathbb{1}^{(2)}
\end{aligned}$$

The similar $1 \to 2$ case follows trivially from replacign the subscripts $1 \to 2$ and this makes it such that the non-identity operators are in the second vector basis and all the operators acting in the first vector basis are identity.

(4) Show that

$$(\Omega_1^{(1)\otimes(2)} + \Omega_2^{(1)\otimes(2)})^2 = (\Omega_1^2)^{(1)} \otimes \mathbb{1}^{(2)} + \mathbb{1}^{(1)} \otimes (\Omega_2^2)^{(2)} + 2\Omega_1^{(1)} \otimes \Omega_2^{(2)}$$

Starting from the LHS, we distribute and write out the operators written in $(1) \otimes (2)$ space as the direct product of two operators written in (1) and (2) respectively.

$$\begin{aligned}
(\Omega_1^{(1)\otimes(2)} + \Omega_2^{(1)\otimes(2)})^2 &= (\Omega_1^{(1)\otimes(2)})^2 + \Omega_1^{(1)\otimes(2)}\Omega_2^{(1)\otimes(2)} + \Omega_2^{(1)\otimes(2)}\Omega_1^{(1)\otimes(2)} + (\Omega_2^{(1)\otimes(2)})^2 \\
&= (\Omega_1^2)^{(1)} \otimes \mathbb{1}^{(2)} + \mathbb{1}^{(1)} \otimes (\Omega_2^2)^{(2)} + 2\Omega_1^{(1)} \otimes \Omega_2^{(2)}
\end{aligned}$$

Exercise 10.1.2 Imagine a fictitious world in which the single-particle Hilbert space is two-dimensional. Let us denote the basis vectors by $|+\rangle$ and $|-\rangle$. Let

$$\sigma_1^{(1)} = \begin{bmatrix} a & b \\ c & d \end{bmatrix} \qquad \text{and} \qquad \sigma_2^{(2)} = \begin{bmatrix} e & f \\ g & h \end{bmatrix}$$

be operators in $\mathbb{V}_1$ and $\mathbb{V}_2$, respectively (the columns and rows have basis vectors + then − in that order such that matrix element $b = \langle + | \sigma_1^{(1)} | - \rangle$ for example. The space $\mathbb{V}_1 \otimes \mathbb{V}_2$ is spanned by vectors $|M_{\pm\pm}\rangle^{(1)\otimes(2)}$ from the set $M = \{|+\rangle\otimes|+\rangle, |+\rangle\otimes|-\rangle, |-\rangle\otimes|+\rangle, |-\rangle\otimes|-\rangle\}$. Show, using the method of images or otherwise, the complete matrix elements of $\sigma_1^{(1)\otimes(2)}$.

(1) We know from the math introduction of this textbook that the matrix elements of $\sigma_1^{(1)\otimes(2)}$ in the $|M_i\rangle$ and $|M_j\rangle$ basis can be given by $\sigma_{1ij}^{(1)\otimes(2)} = \langle M_i | \sigma_1^{(1)\otimes(2)} | M_j \rangle$, or equivalently, the elements of the jth column are simply the components of the jth basis vector transformed by the operator. So thus the columns of $\sigma_1^{(1)\otimes(2)}$ take the form $\sigma_1^{(1)\otimes(2)} |M_{\pm\pm}\rangle^{(1)\otimes(2)}$. We do this for each basis vector making use of $\sigma_1^{(1)} |+\rangle$ and $\sigma_1^{(1)} |-\rangle$ which can be calculated easily from matrix multiplication.

$$\begin{aligned}
\sigma_1^{(1)\otimes(2)} |M_{++}\rangle^{(1)\otimes(2)} &= (\sigma_1^{(1)} \otimes \mathbb{1}^{(2)})(|+\rangle \otimes |+\rangle) \\
&= \sigma_1^{(1)} |+\rangle \otimes |+\rangle \\
&= (a|+\rangle + c|-\rangle) \otimes |+\rangle \\
&= a|+\rangle \otimes |+\rangle + c|-\rangle \otimes |+\rangle
\end{aligned}$$

$$\begin{aligned}
\sigma_1^{(1)\otimes(2)} |M_{+-}\rangle^{(1)\otimes(2)} &= (\sigma_1^{(1)} \otimes \mathbb{1}^{(2)})(|+\rangle \otimes |-\rangle) \\
&= \sigma_1^{(1)} |+\rangle \otimes |-\rangle \\
&= (a|+\rangle + c|-\rangle) \otimes |-\rangle \\
&= a|+\rangle \otimes |-\rangle + c|-\rangle \otimes |-\rangle
\end{aligned}$$

$$\begin{aligned}
\sigma_1^{(1)\otimes(2)} |M_{-+}\rangle^{(1)\otimes(2)} &= (\sigma_1^{(1)} \otimes \mathbb{1}^{(2)})(|-\rangle \otimes |+\rangle) \\
&= \sigma_1^{(1)} |-\rangle \otimes |+\rangle \\
&= (b|+\rangle + d|-\rangle) \otimes |+\rangle \\
&= b|+\rangle \otimes |+\rangle + d|-\rangle \otimes |+\rangle
\end{aligned}$$

$$\begin{aligned}
\sigma_1^{(1)\otimes(2)} |M_{--}\rangle^{(1)\otimes(2)} &= (\sigma_1^{(1)} \otimes \mathbb{1}^{(2)})(|-\rangle \otimes |-\rangle) \\
&= \sigma_1^{(1)} |-\rangle \otimes |-\rangle \\
&= (b|+\rangle + d|-\rangle) \otimes |-\rangle \\
&= b|+\rangle \otimes |-\rangle + d|-\rangle \otimes |-\rangle
\end{aligned}$$

This results in the final matrix expression

$$\sigma_1^{(1)\otimes(2)} \hat{=} \begin{bmatrix} a & 0 & b & 0 \\ 0 & a & 0 & b \\ c & 0 & d & 0 \\ 0 & c & 0 & d \end{bmatrix}$$

(2) Similarly, $\sigma_2^{(1)\otimes(2)}$ can be calculated too.

$$\begin{aligned}
\sigma_2^{(1)\otimes(2)} |M_{++}\rangle^{(1)\otimes(2)} &= |+\rangle \otimes \sigma_2^{(2)} |+\rangle = e\,|+\rangle \otimes |+\rangle + g\,|+\rangle \otimes |-\rangle = e\,|M_{++}\rangle + g\,|M_{+-}\rangle \\
\sigma_2^{(1)\otimes(2)} |M_{+-}\rangle^{(1)\otimes(2)} &= |+\rangle \otimes \sigma_2^{(2)} |-\rangle = f\,|+\rangle \otimes |+\rangle + h\,|+\rangle \otimes |-\rangle = f\,|M_{++}\rangle + h\,|M_{+-}\rangle \\
\sigma_2^{(1)\otimes(2)} |M_{-+}\rangle^{(1)\otimes(2)} &= |-\rangle \otimes \sigma_2^{(2)} |+\rangle = e\,|-\rangle \otimes |+\rangle + g\,|-\rangle \otimes |-\rangle = e\,|M_{-+}\rangle + g\,|M_{--}\rangle \\
\sigma_2^{(1)\otimes(2)} |M_{--}\rangle^{(1)\otimes(2)} &= |-\rangle \otimes \sigma_2^{(2)} |-\rangle = f\,|-\rangle \otimes |+\rangle + h\,|-\rangle \otimes |-\rangle = f\,|M_{-+}\rangle + h\,|M_{--}\rangle
\end{aligned}$$

Which constructs the correct representation

$$\sigma_2^{(1)\otimes(2)} \hat{=} \begin{bmatrix} e & f & 0 & 0 \\ g & h & 0 & 0 \\ 0 & 0 & e & f \\ 0 & 0 & g & h \end{bmatrix}$$

(3) We then try computing $(\sigma_1\sigma_2)^{(1)\otimes(2)}$ first by taking the matrix product of $\sigma_1^{(1)\otimes(2)}$ and $\sigma_2^{(1)\otimes(2)}$, and also try directly computing the matrix elements. Firstly by matrix multiplication:

$$\sigma_1^{(1)\otimes(2)}\sigma_2^{(1)\otimes(2)} = \begin{bmatrix} a & 0 & b & 0 \\ 0 & a & 0 & b \\ c & 0 & d & 0 \\ 0 & c & 0 & d \end{bmatrix} \begin{bmatrix} e & f & 0 & 0 \\ g & h & 0 & 0 \\ 0 & 0 & e & f \\ 0 & 0 & g & h \end{bmatrix} = \begin{bmatrix} ae & af & be & bf \\ ag & ah & bg & bh \\ ce & cf & de & df \\ cg & ch & dg & dh \end{bmatrix}$$

Then computing each element: We will spare the repeated task of explicitly computing every single one and get as far as we can with a general statement. We will make use of basis vectors $|M_a\rangle^{(1)\otimes(2)}, |M_b\rangle^{(1)\otimes(2)} \in M$. We will also use the notation that $|M_a\rangle = |M_a\rangle^{(1)} \otimes |M_a\rangle^{(2)}$. Then,

$$\begin{aligned}
(\sigma_1\sigma_2)^{(1)\otimes(2)} &= \sum_{a,b} \langle M_a| \left(\sigma_1^{(1)} \otimes \sigma_2^{(2)}\right) |M_b\rangle\, |M_a\rangle \langle M_b| \\
&= \sum_{a,b} \langle M_a^{(1)}|\, \sigma_1^{(1)}\, |M_b^{(1)}\rangle \langle M_a^{(2)}|\, \sigma_2^{(2)}\, |M_b^{(2)}\rangle\, |M_a\rangle \langle M_b|
\end{aligned}$$

Notice that this means that there will be two scalar quantities for each matrix element of $(\sigma_1\sigma_2)^{(1)\otimes(2)}$. One scalar comes from $\langle M_a^{(1)}|\, \sigma_1^{(1)}\, |M_b^{(1)}\rangle$ which projects $\sigma_1^{(1)}$ on to the basis vector contained in $\mathbb{V}_1$ of the row and column. The second scalar comes from $\langle M_a^{(2)}|\, \sigma_2^{(2)}\, |M_b^{(2)}\rangle$ which projects $\sigma_2^{(2)}$ on to the basis vector contained in $\mathbb{V}_2$ of the row and column. This can easily be verified to agree with all the matrix elements found when calculating it via matrix multiplication.

Exercise 10.1.3 Consider the Hamiltonian of the coupled mass system:

$$\mathcal{H} = \frac{p_1^2}{2m} + \frac{p_2^2}{2m} + \frac{1}{2}m\omega^2[x_1^2 + x_2^2 + (x_1 - x_2)^2]$$

We know from Example 1.8.6 that $\mathcal{H}$ can be decoupled if we use normal coordinates

$$x_{I,II} = \frac{x_1 \pm x_2}{2^{1/2}}$$

and the corresponding momenta

$$p_{I,II} = \frac{p_1 \pm p_2}{2^{1/2}}$$

(1) We rewrite the Hamiltonian in normal coordinates via simple substitution. Note that $x_1^2 + x_2^2 = x_I^2 + x_{II}^2$ and same for the momentum conjugate.

$$\mathcal{H} = \frac{1}{2m}(p_I^2 + p_{II}^2) + \frac{1}{2}m\omega^2[x_I^2 + 3x_{II}^2]$$

We also verify that the normal coordinates are canonical. Starting from the definition of poisson brackets we evaluate it making note that $\partial x_i/\partial p_j = \partial p_i/\partial x_j = 0$ for any $i, j = I, II$.

$$\begin{aligned}\{x_i, p_j\} &= \sum_{k=1}^{2}\left(\frac{\partial x_i}{\partial x_k}\frac{\partial p_j}{p_k} - \frac{\partial x_i}{\partial p_k}\frac{\partial p_j}{\partial x_k}\right) = \frac{\partial x_i}{\partial x_1}\frac{\partial p_j}{\partial p_1} + \frac{\partial x_i}{\partial x_2}\frac{\partial p_j}{\partial p_2} \\ &= \frac{1}{2} + (2\delta_{ij} - 1)\frac{1}{2} = \delta_{ij}\end{aligned}$$

Now we quantize the system, promoting these variables to operators. We can naively assume that these operators in the normal coordinate basis can be represented in the x-basis as $X_i \to x_i$ and $P_j \to -i\hbar\partial/\partial x_j$. By evaluating the commutator with a test function $f = f(x_I, x_{II})$ we can verify this choice of representation satisfies the canonical commutation relation.

$$[X_i, P_j]f = -x_i i\hbar\frac{\partial f}{\partial x_j} + i\hbar\frac{\partial}{\partial x_j}(x_i f) = i\hbar\delta_{ij}f$$

Promoting the hamiltonian leads to

$$H = \frac{1}{2m}(P_I^2 + P_{II}^2) + \frac{1}{2}m\omega^2(X_I^2 + 3X_{II}^2)$$

Lastly, we write the eigenvalue equation for H in the simultaneous eigenbasis of X_I and X_{II}. For that we define $X_i |X_I, X_{II}\rangle = X_i |X_I, X_{II}\rangle$ for $i = I, II$. We then write the eigenvalue equation for H in the energy basis first, then project it onto the $|X_I, X_{II}\rangle$ basis.

$$\begin{aligned}H|\psi\rangle = E|\psi\rangle \to \langle X_I, X_{II}|H|\psi\rangle &= \langle X_I, X_{II}|E|\psi\rangle \\ &= -\frac{\hbar^2}{2m}\left(\frac{\partial^2}{\partial x_I^2} + \frac{\partial^2}{\partial x_{II}^2}\right)\psi(x_I, x_{II}) + \frac{1}{2}m\omega^2(x_I^2 + 3x_{II}^2)\psi(x_I, x_{II}) \\ &= E\psi(x_I, x_{II})\end{aligned}$$

(2) Quantize the system directly, by promoting x_1, x_2, p_1 and p_2 to quantum operators. Write the eigenvalue equation for H in the simultaneous eigenbasis of X_1 and X_2. Now

change from x_1, x_2 (and of course $\partial/\partial x_1$, $\partial/\partial x_2$) to x_I, x_{II} (and $\partial/\partial x_I$, $\partial/\partial x_{II}$) in the differential equation. You should end up with the result from part (1)

We promote H

$$H = \frac{P_1^2}{2m} + \frac{P_2^2}{2m} + \frac{1}{2}m\omega^2[X_1^2 + X_2^2 + (X_1 - X_2)^2]$$

and write the eigenvalue equation in the simultaneous eigenbasis. We will make use of the chain rule in order to swap from derivatives in the canonical coordinates to derivatives in normal coordinates. It is such that

$$\frac{\partial}{\partial x_1} = \frac{1}{\sqrt{2}}\left(\frac{\partial}{\partial x_I} + \frac{\partial}{\partial x_{II}}\right)$$

and the same for a derivative over x_2 except the second term is negative. In fact, we demonstrate how the negative allows for the cross terms to cancel when going from derivatives over canonical coordinates to derivatives over normal coordinates.

$$\begin{aligned}
&\langle X_1, X_2| H |\psi\rangle = \langle X_1, X_2| E |\psi\rangle \\
&= \left(-\frac{\hbar^2}{2m}\left(\frac{\partial^2}{\partial x_1^2} + \frac{\partial^2}{\partial x_2^2}\right) + \frac{1}{2}m\omega^2(x_1^2 + x_2^2 + (x_1 - x_2)^2)\right)\psi(x_1, x_2) \\
&= \left(-\frac{\hbar^2}{2m}\left(\frac{\partial^2}{\partial x_I^2} + \frac{\partial^2}{\partial x_{II}^2}\right) + \frac{1}{2}m\omega^2(x_I^2 + 3x_{II}^2)\right)\psi(x_1, x_2) = E\psi(x_1, x_2)
\end{aligned}$$

After reminding ourselves that x_1, x_2 are just functions of x_I, x_{II}, then we recover the same expression.

Exercise 10.2.1 (Particle in a 3-Dimensional Box), Recall that a particle in a one-dimensional box extending from x=0 to L is confined to the region $0 \leq x \leq L$; its wave function vanishes at the edges $x = 0$ and L and beyond (Exercise 5.2.5). Consider now a particle confined in a three-dimensional cubic box of volume L^3. Choosing as the origin one of its corners, and the x, y, and z axes along the three edges meeting there, show that the normalized energy eigenfunctions are.

$$\psi_E(x, y, z) = \left(\frac{2}{L}\right)^{1/2} \sin\left(\frac{n_x \pi x}{L}\right)\left(\frac{2}{L}\right)^{1/2} \sin\left(\frac{n_y \pi y}{L}\right)\left(\frac{2}{L}\right)^{1/2} \sin\left(\frac{n_z \pi z}{L}\right)$$

where

$$E = \frac{\hbar^2\pi^2}{2ML^2}(n_x^2 + n_y^2 + n_z^2)$$

and n_i are positive integers.

This problem is trivial by remembering that in 3D since cross-commutator's are zero, the momentum terms in the Hamiltonian are all separable and the potential terms can all be separated such that the 3D Hamiltonian can be $H = H_x + H_y + H_z$. If my Hamiltonian is separable then we can treat them as three separate differential equations that correspond to separable energies $E_{tot} = E_x + E_y + E_z$. Then each of these individual equations are identical to the single dimension particle in a box problem. The equation above is merely $\psi_E(x, y, z) = \psi_E(x)\psi_E(y)\psi_E(z)$ where each ψ is the generic solution of the 1D particle in a box making substitutions for each coordinates. The energy term also directly follows from the generic 1D particle in a box energy.

Exercise 10.2.2 Quantize the 2D oscillator for which

$$\mathcal{H} = \frac{p_x^2 + p_y^2}{2m} + \frac{1}{2}m\omega_x^2 x^2 + \frac{1}{2}m\omega_y^2 y^2$$

(1) Show that the allowed energies are

$$E = (n_x + 1/2)\hbar\omega + (n_y + 1/2)\hbar\omega_y, \quad n_x, n_y = 0, 1, 2...$$

Once again, the hamiltonian is promoted in such a way that it is seperable $H = H_x + H_y$. Then $[x, y] = 0$ implies that I can construct a simultaneous eigenket $|x, y\rangle$ to project the energy eigenvalue equation on to, revealing that with the ansatz $\psi(x, y) = \psi(x)\psi(y)$ and $E_{tot} = E_x + E_y$, then the energies are just the sum of 1D-oscillator energy on two different coordinates x and y.

(2) Writing down the corresponding wave function in terms of single oscillator wave functions. Verify that they have definite parity (even/odd) number $x \to -x$, $y \to -y$ and that the parity depends only on $n = n_x + n_y$.

Using the ansatz mentioned earlier, we simply multiply the 1D oscillator for both coordinates giving us

$$\psi(x, y) = \psi(x)\psi(y) = A_n \exp(-m\omega x^2/2\hbar) H_{n_x}((m\omega/\hbar)^{1/2}x) \exp(-m\omega y^2/2\hbar) H_{n_y}((m\omega/\hbar)^{1/2}y)$$

The parity is determined by the Hermite polynomials. Since the exponentials are always even functions the parity of $n = n_x + n_y$ and H_{n_x}, H_{n_y} fully defines the total parity. if n_x and n_y have the same parity then the resulting function is even. If they have different parity then the resulting function is odd.

(3) Consider next the isotropic oscillator ($\omega_x = \omega_y$). Write explicit, normalized eigenfunctions of the first three states (that is, for the cases $n = 0$ and 1). Reexpress your results in terms of polar coordinates ρ, and ϕ (for later use). Show that the degeneracy of a level with $E = (n + 1)\hbar\omega$ is $(n + 1)$.

For $n = 0$, $H_0 = 1$. So

$$\psi_0(x, y) = A_n e^{-m\omega(x^2+y^2)/2\hbar}$$

The coefficient is determined through normalization

$$1 = \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} dxdy|\psi_0|^2 = |A_n|^2 \int_0^{\infty}\int_0^{2\pi} re^{-m\omega r^2/2\hbar} drd\theta = |A_n|^2 \frac{\pi\hbar}{m\omega}$$

So

$$\psi_0(x, y) = \sqrt{\frac{m\omega}{\pi\hbar}} e^{-m\omega(x^2+y^2)/2\hbar}$$

For $n = 1$, $H_1(x) = 2x$

So

$$\psi(x, y) = A_n e^{-m\omega(x^2+y^2)/2\hbar} \left(2\sqrt{\frac{m\omega}{\hbar}}x\right)$$

In the same procedure as before

$$1 = \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} dxdy|A_n|^2 4\frac{m\omega}{\hbar}x^2 e^{-m\omega x^2/\hbar}e^{-m\omega y^2/\hbar} = |A_n|^2 2\frac{\pi\hbar}{m\omega}$$

Thus $|A_n| = \frac{m\omega}{2\pi\hbar}$ and

$$\psi_1(x,y) = \sqrt{2}\frac{m\omega}{\hbar}xe^{-m\omega(x^2+y^2)/2\hbar}$$

for the case where $n = n_x = 1$ and for the $n = n_y = 1$ degenerate case, the x outside of the exponential is swapped with a y

Exercise 10.2.3 Quantize the 3D isotropic oscillator for which

$$\mathcal{H} = \frac{p_x^2 + p_y^2 + p_z^2}{2m} + \frac{1}{2}m\omega^2(x^2+y^2+z^2)$$

(1) The energy is $E = (n + 3/2)\hbar\omega$; $n = n_x + n_y + n_z$; $n_x, n_y, n_z = 0, 1, 2....$ This is easily seen from the Hamiltonian being separable leading to separable Schrodinger equations and wave functions. The expression for energy is merely the sum of three independent quantum harmonic oscillators.

(2) The corresponding eigenfunctions can be written in terms of single-oscillator wave functions where $\alpha = (m\omega/\hbar)^{1/2}$

$$\begin{aligned}\psi_n(x,y,z) &= A_n\psi(x)\psi(y)\psi(z)\\ &= A_n e^{-m\omega x^2/2\hbar}H_{n_x}(\alpha x)e^{-m\omega y^2/2\hbar}H_{n_y}(\alpha y)e^{-m\omega z^2/2\hbar}H_{n_z}(\alpha z)\end{aligned}$$

The wave function's parity entirely depends on n. As H_0 is even, H_1 is odd, and alternating so forth. This means that if n is even, then either all the coordinate n_i $i \in 1, 2, 3$ are even, or one is even and two are odd. In the former case, then all Hermite polynomials are even. In the latter case then all the Hermite polynomials multiplied together are even. The same logic goes for if n is odd. Either they are all odd which generates three Hermite polynomials that are all together odd, or they are two even polynomials multiplied by an odd polynomials, resulting in a totally odd function. In other words, the parity of any ψ_n is $(-1)^n$.

We can write explicit examples for the first four states written in spherical coordinates. We omit the calculation of the normalization constant A_n but it is just a typical integration over all space in radial coordinates in order to deduce it's value.

The first state has n=0.

$$\psi_1(r) = A_n e^{-m\omega r^2/2\hbar}$$

The next three states are degenerate across three different states since $n = 1$ can imply $\psi_{xyz} = \psi_{100}$, ψ_{010}, or ψ_{001}.

$$\begin{aligned}\psi_{100} &= A_n e^{-m\omega r^2/2\hbar}(2\alpha x)\\ \psi_{010} &= A_n e^{-m\omega r^2/2\hbar}(2\alpha y)\\ \psi_{001} &= A_n e^{-m\omega r^2/2\hbar}(2\alpha z)\end{aligned}$$

Make the appropriate spherical substitution $x = r\sin\phi\cos\theta$,$y = r\sin\phi\sin\theta$, and $z = r\cos\phi$ in order to convert the above into entirely spherical coordinates.

Exercise 10.3.1 Two identical bosons are found to be in states $|\phi\rangle$ and $|\psi\rangle$. Write down the normalized state vector describing the system when $\langle\phi|\psi\rangle \neq 0$.

Bosons have symmetrical state vectors, thus construct the statevector by taking the direct product $|\phi\rangle \otimes |\psi\rangle \equiv |\phi\psi\rangle$ and permuting it under $\psi \longleftrightarrow \phi$. We sum these two states and normalize it to yield the normalize statevector

$$|\phi, \psi; S\rangle = A_n(|\phi\psi\rangle + |\psi\phi\rangle)$$

Since $\langle\phi|\psi\rangle \neq 0$, we don't have a definite normalization vector but as far as we can get is calculated below.

$$\begin{aligned}\langle\phi, \psi; S|\phi, \psi; S\rangle =& |A_n|^2(\langle\phi\psi|\phi\psi\rangle + \langle\phi\psi|\psi\phi\rangle + \langle\psi\phi|\phi\psi\rangle + \langle\psi\phi|\psi\phi\rangle) = 1 \\ & |A_n|^2 = 1/(2 + |\langle\psi|\phi\rangle|^2)\end{aligned}$$

Exercise 10.3.2 When an energy measurement is made on a system of three bosons in a box, the n values obtained were $3, 3$ and 4. Write down a symmetrized, normalized state vector.

There are three states that could have produced this result, and we normalize those three to produce

$$|3, 3, 4; S\rangle = \frac{1}{\sqrt{3}}(|334\rangle + |343\rangle + |433\rangle)$$

Exercise 10.3.3 Imagine a situation in which there are three particles and only three states a, b, and c available to them. Show that the number of allowed, distinct configurations for this system is.

(1) 27 if they are labeled. - If the particles are labeled, then they are distinguishable and the unique states just amounts to the total permutations that three particles can take on the values a,b, or c. This amounts to $3 * 3 * 3$ states which equals 27.

(2) 10 If they are bosons. - If these particles are bosons then the states are unique up to exchange of particles. We can treat this as the three particles can either be all the same, two the same and one different, or all different. Then we use combinatorics to represents all the possibilities for each of cases.

$$\binom{3}{1} + \binom{3}{1}\binom{2}{1} + \binom{3}{3} = 10$$

(3) 1 if they are fermions. For fermions every particle must have a different state. So for three particles that can occupy three states, there is only one distinct configuration.

Exercise 10.3.4 Two identical particles of mass m are in a 1D box of length L. Energy measurement of the system yields the value $E_{sys} = \hbar^2\pi^2/mL^2$. Write down the state vector of the system. Repeat for $E_{sys} = 5\hbar^2\pi^2/2mL^2$. (There are two possible vectors in this case.) You are not told if they are bosons or fermions. You may assume that the only degrees of freedom are orbital.

Note that the 1D particle in a box has Energy solutions $E_n = \hbar^2\pi^2n^2/2mL^2$. Then this two particle system has a system energy that is a sum of two individual particle's energies $E_{sys} = \hbar^2\pi^2(n_1^2 + n_2^2)/2mL^2$. The first case is only satisfied if $n_1 = n_2 = 1$. Since these particles are only free in it's orbital value, and this case has identical orbital values across both particles, this case is only possible if these are bosons instead of fermions (due to the Pauli exclusion principle). The second case is possible if one of the energy levels is $n = 2$ and the other $n = 1$. There is a degeneracy because you may swap which particle gets which energy level. In this scenario, we are unable to tell if it is bosonic or fermionic.

Exercise 10.3.5 Consider the exchange operator P_{12} whose action on the X basis is

$$P_{12}\left|x_1, x_2\right\rangle = \left|x_2, x_1\right\rangle$$

(1) Show that P_{12} has eigenvalues ± 1 (It is hermitian and unitary).

Let P_{12} have an eigenvector $|\psi\rangle$, then $P_{12}\left|\psi\right\rangle = \lambda\left|\psi\right\rangle$. Note that from the definition of P_{12} action, we can see that $P_{12}^2 = \mathbb{1}$. Combining these two facts we get $P_{12}^2\left|\psi\right\rangle = \lambda^2\left|\psi\right\rangle = \left|\psi\right\rangle$ which shows how eigenvalue $\lambda = \pm 1$.

An alternative method is to show that the operator is unitary and hermitian. We have already noticed that $P_{12} = \mathbb{1}$ which implies $P = P^{-1}$. We can also observe that P_{12} preserves inner products which implies it is unitary and hermitian. We let $P_{12}\left|y_1, y_2\right\rangle = \left|y_2, y_1\right\rangle$

$$\langle x_1, x_2|y_1, y_2\rangle = \delta_{x_1,y_1}\delta_{x_2,y_2} = \langle x_2, x_1|y_2, y_1\rangle = \langle x_1, x_2|\, P^\dagger P\left|y_1, y_2\right\rangle$$

Since P is unitary and now hermitian, it's eigenvalues must have unit modulus and are real. The only two options at $\lambda = \pm 1$

(2) Show that its action on the basis ket $|\omega_1, \omega_2\rangle$ is also to exchange the labels 1 and 2 and hence that $\mathbb{V}_{S/A}$ are its eigenspaces with eigenvalues ± 1.

We show it's action on this basis ket by expanding it in the $|x_2, x_1\rangle$ basis.

$$\begin{aligned} P\left|\omega_1, \omega_2\right\rangle &= \int dx_2 \int dx_1 P\left|x_2, x_1\right\rangle \left\langle x_2, x_1|\omega_1, \omega_2\right\rangle \\ &= \int dx_2 \int dx_1 \left|x_1, x_2\right\rangle \omega_2(x_1)\omega_1(x_2) \\ &= \int dx_2 \int dx_1 \left|x_1, x_2\right\rangle \left\langle x_1, x_2|\omega_2, \omega_1\right\rangle = \left|\omega_2, \omega_1\right\rangle \end{aligned}$$

We can see that $\mathbb{V}_{S/A}$ is its eigenspace because a basis of that eigenspace is given by vectors of the form $|\omega_1, \omega_2\rangle \pm |\omega_2, \omega_1\rangle$. If you act P_{12} on these two vectors we arrive at eigenvalues of ± 1.

(3) Show that $P_{12}X_1P_{12} = X_2, \quad P_{12}X_2P_{12} = X_1$ and similarly for P_1 and P_2. Then show that $P_{12}\Omega(X_1, P_1; X_2, P_2)P_{12} = \Omega(X_2, P_2; X_1, P_1)$. [Consider the action on $|x_1, x_2\rangle$ or $|p_1, p_2\rangle$. As for the functions of X and P, assume they are given by power series and consider any term in the series. If you need help, peek into the discussion leading to Eq. 11.2.22

We consider the action of $P_{12}X_1P_{12}$ on $|x_1, x_2\rangle$

$$P_{12}X_1P_{12}\left|x_1, x_2\right\rangle = P_{12}X_1\left|x_2, x_1\right\rangle = x_2\left|x_1, x_2\right\rangle = X_2\left|x_1, x_2\right\rangle$$

Multiplying the top equation on both sides by P_{12} also yields the $P_{12}X_2P_{12} = X_1$ variant.

the P_1 and P_2 versions of the equation follows exactly the same since we can use it's action on $|p_1, p_2\rangle$ and P_{12} will swap the labels none the same (as seen in part 2).

For the arbitrary Ω operator, we can use it's eigenkets $|\omega_1, \omega_2\rangle$. Moreover, because $[X_i, P_j] = i\hbar\delta_{ij}$ (commutes to a scalar quantity or nothing), then we can choose a standard order of the parameters of Ω and expand it as a generic power series

$$\Omega(X_1, P_1; X_2, P_2) = \sum_{i,j,k,l} c_{i,j,k,l} X_1^i P_1^j X_2^k P_2^l$$

Earlier we have identities that when raised to a power yield $X_2^i = P_{12} X_1^i P_{12}$ and the same for other similar equations. So now we can observe that

$$\begin{aligned} P_{12}\Omega(X_1, P_1; X_2, P_2)P_{12} &= \sum_{i,j,k,l} c_{i,j,k,l}(P_{12}X_1^i P_{12})(P_{12}P_1^j P_{12})(P_{12}X_2^k P_{12})(P_{12}P_2^l P_{12}) \\ &= P_{12}\Omega(X_2, P_2; X_1, P_1)P_{12} \end{aligned}$$

(4) Show that the Hamiltonian and propagator for two identical particles are left unaffected under $H \to P_{12}HP_{12}$ and $U \to P_{12}UP_{12}$. Given this, show that any eigenstate of P_{12} continues to remain an eigenstate with the same eigenvalue as time passes, i.e., elements of $\mathbb{V}_{S/A}$ never leave the symmetric or antisymmetric subspaces they start in.

We know that H and U is a function of $X_1, P_1; X_2, P_2$. Given part 3 above, the transformation of H and U under P_{12} on both sides amounts to just swapping labels for identical particles. Since they are identical and it is just swapping models, there is no physical difference. For U the swapping of labels for the evolution of two identical particles also would amount to no physical change.

Let $|\psi\rangle$ be the eigenstate of P_{12} at $t = 0$ s.t. $P_{12}|\psi\rangle = \lambda|\psi\rangle$ where $\lambda = \pm 1$. We can multiply both sides by $(P_{12}U(t))$ in order to get $P_{12}U(t)P_{12}|\psi\rangle = P_{12}U(t)\lambda|\psi\rangle$. Using the substitution of U on the LHS and noticing that $U(t)|\psi\rangle = |\psi(t)\rangle$, we get $U(t)|\psi\rangle = P_{12}\lambda|\psi(t)\rangle \to P_{12}|\psi(t)\rangle = \lambda|\psi(t)\rangle$. This eigenvalue equation shows that the eigenstate of P_{12} continues to remain an eigenstate with the same eigenvalue as time passes.

Exercise 10.3.6 Consider a composite object such as the hydrogen atom. Will it behave as a boson or fermion? Argue in general that objects containing an even/odd number of fermions will behave as bosons/fermions.

We will consider two composite particles comprised on n fermions

$$|A\rangle = \frac{1}{(n!)^{1/2}}\left(\sum_{a_+ \in S_A^+} |a_+\rangle - \sum_{a_- \in S_A^-} |a_-\rangle\right)$$

and

$$|B\rangle = \frac{1}{(n!)^{1/2}}\left(\sum_{b_+ \in S_B^+} |b_+\rangle - \sum_{b_- \in S_B^-} |b_-\rangle\right)$$

where $S_A^{\pm}$ is the set of all even/odd permutations of $a_1, a_2, a_3 ... a_n$ (the same for $S_B^{\pm}$).

The combined statevector of the two composite particles is $|A\rangle \otimes |B\rangle$.

$$|A\rangle \otimes |B\rangle = \frac{1}{(n!)}\left(\sum_{a_+,b_+} |a_+b_+\rangle + \sum_{a_-,b_-} |a_-b_-\rangle - \sum_{a_+,b_-} |a_+b_-\rangle - \sum_{a_-b_+} |a_-b_+\rangle\right)$$

The determination of whether the composite system is fermionic or bosonic is predicated on whether $|A\rangle \otimes |B\rangle$ is symmetric or antisymmetric under the interchange of particles, i.e. $a_i \leftrightarrow b_i$

$$|B\rangle \otimes |A\rangle = \frac{1}{(n!)}\Big(\sum_{a_+,b_+} |b_+ a_+\rangle + \sum_{a_-,b_-} |b_- a_-\rangle - \sum_{a_+,b_-} |b_+ a_-\rangle - \sum_{a_- b_+} |b_- a_+\rangle \Big)$$

Idk how to do this problem. The easiest solution is to just say that fermions have half integer spin and bosons have integer. So even numbers of fermions sum the spin to make integer spins (bosons). but odd fermions make it still integer half composite particles.